

10.18

a.

Percentage of mothers with some college education: 12.14953 %

Percentage of fathers with some college education: 11.68224 %

b.

Yes, the magnitudes are important. Strong correlations are required to satisfy the "relevance" assumption of an instrumental variable (IV) and avoid the severe biases caused by weak instruments. Dummy variables capture a "threshold effect" and reduce measurement error, as respondents are more likely to accurately remember if their parents went to college than their exact years of schooling.

c.

95% interval estimation for the coefficient of educ is [-0.001219763, 0.1532557].

d.

The F-statistic for the hypothesis that MOTHERCOLL has no effect on EDUC is **63.56**. According to the standard econometric rule of thumb, an instrumental variable is considered strong if the first-stage F-statistic is greater than 10. Since 63.56 is substantially larger than 10, MOTHERCOLL is a very strong instrument for EDUC.

e.

95% interval estimation for the coefficient of educ is [0.02751845, 0.1481769].

The 95% interval is narrower than the one in part (c). Adding a second valid and relevant instrumental variable increases the efficiency of the 2SLS estimator. By capturing more exogenous variation in EDUC from both parents rather than just the mother, the standard error of the estimated coefficient decreases, resulting in a tighter, more precise confidence interval.

f.

Joint F-statistic is 56.963

Because the joint F-statistic is greater than 10, reject the null hypothesis and conclude that the instruments are jointly significant and adequately strong.

g.

The p-value for the Sargan test is 0.626

Because the p-value is greater than 0.05, we fail to reject the null hypothesis, indicating that the instruments are valid and appropriately excluded from the main wage equation.

10.20

a.

Microsoft stock is classified as risky relative to the market portfolio over this period because the estimate (1.201840) is strictly greater than 1.

b.

The p-value for the RANK coefficient is $< 2e-16$. This indicates it is highly statistically significant at any standard level. Multiple R-squared is **0.9126**. This means that RANK explains about 91.26% of the variance in the market risk premium. The rule of thumb is that an instrument is strong if the first-stage F-statistic is greater than 10. Our F-statistic is **1858**, which vastly exceeds this threshold. So, it is a very strong IV.

c.

The p-value for the v_{hat} coefficient is **0.0428**

Because the p-value is **greater than 0.01**, we fail to reject the null hypothesis and conclude that the market return is **exogenous** at the 1% significance level.

d.

Yes. If the market return is measured with error, the OLS estimate suffers from **attenuation bias**, meaning it is artificially pulled closer to zero. Therefore, we should expect the IV estimate to be **larger** ($1.278318 > 1.201840$) than the OLS estimate, as the IV method corrects for this downward bias.

e.

The F-statistic is **951.3** with a p-value of $< 2.2e-16$. Because this p-value is practically zero, we reject the null hypothesis and conclude that the instruments (RANK and POS) are jointly highly significant. The multiple R-squared of this first-stage regression is **0.9149**. This means the instruments explain about 91.49% of the variation in the market risk premium.

f.

The p-value for the $\hat{v}_e$ coefficient is 0.0287. Because the p-value is **greater than 0.01**, we fail to reject the null hypothesis and can conclude that the market return is **exogenous** at the 1% level of significance.

g.

Yes. If the market return is measured with error, standard OLS suffers from *attenuation bias*, meaning the OLS coefficient is pushed artificially closer to zero. By using valid instrumental variables, 2SLS corrects for this measurement error. Therefore, we should expect the IV estimate of the beta to be **larger** ($1.283118 > 1.201840$) than the OLS estimate.

h.

The calculated Sargan test statistic is **0.558** with a p-value of **0.455**. The null hypothesis of the Sargan test is that all overidentifying instruments are valid (i.e., they are exogenous and uncorrelated with the regression error term). Because the p-value (0.455) is significantly greater than the 0.05 (5%) significance level, we **fail to reject** the null hypothesis, indicating that our surplus instrumental variables (RANK and POS) are indeed **valid**. They satisfy the necessary exogeneity condition and are appropriately used in our IV/2SLS estimation.